Marketing Revenue and Conversion Forecasting (Excel)

Personal Project

Project Overview

This project focuses on forecasting key marketing performance metrics to support planning, budgeting, and strategic decision-making. Using 24 months of historical marketing data, I built multiple Excel-based forecasting models to estimate future revenue, conversions, and performance trends while accounting for growth patterns and seasonality.

The goal was not only to predict future performance, but to translate forecasts into clear, actionable insights for non-technical stakeholders.

Business Problem

The marketing team needed reliable short- and medium-term forecasts to:

- Set realistic revenue and conversion targets
- Plan marketing budgets effectively
- Understand the impact of ad spend on revenue
- Prepare for seasonal demand fluctuations

Historical performance data existed, but forecasts were not standardized or clearly explained, limiting their usefulness for decision-making.

Dataset

The dataset contains monthly marketing performance data from January 2023 to December 2024, including:

- Website Sessions
- Ad Spend (€)
- Leads
- Conversions
- Revenue (€)

The data reflects realistic growth trends and seasonal peaks typical of digital marketing environments.

Approach & Methodology

1. Data Preparation

- Validated data completeness and consistency
- Converted month values to date format
- Created a time index to support trend-based forecasting

2. Forecasting Methods Used

Linear Trend Forecasting

- Used Excel's FORECAST.LINEAR function
- Applied to revenue, conversions, and ad spend
- Best suited for long-term growth trends

Moving Average Forecasting

- Used 3-month moving averages
- Smoothed short-term volatility
- Used for near-term planning

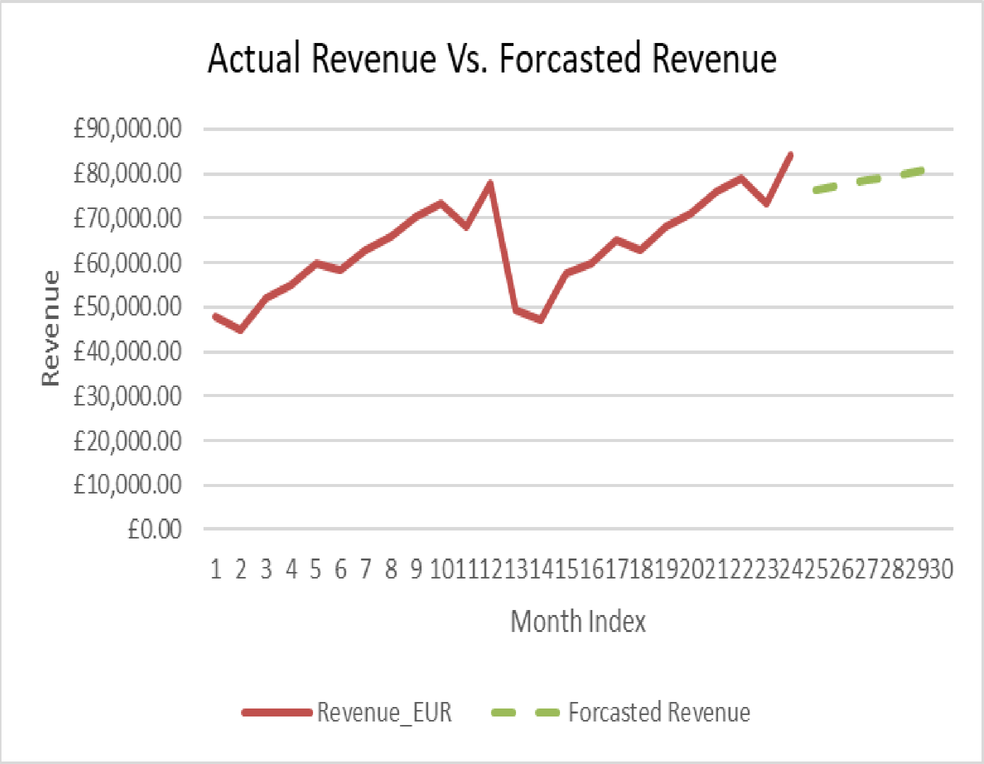
Seasonality-Aware Forecasting (ETS)

- Used Excel's Forecast Sheet (ETS)
- Captured recurring seasonal patterns
- Particularly effective for revenue forecasting

Key Analysis & Insights

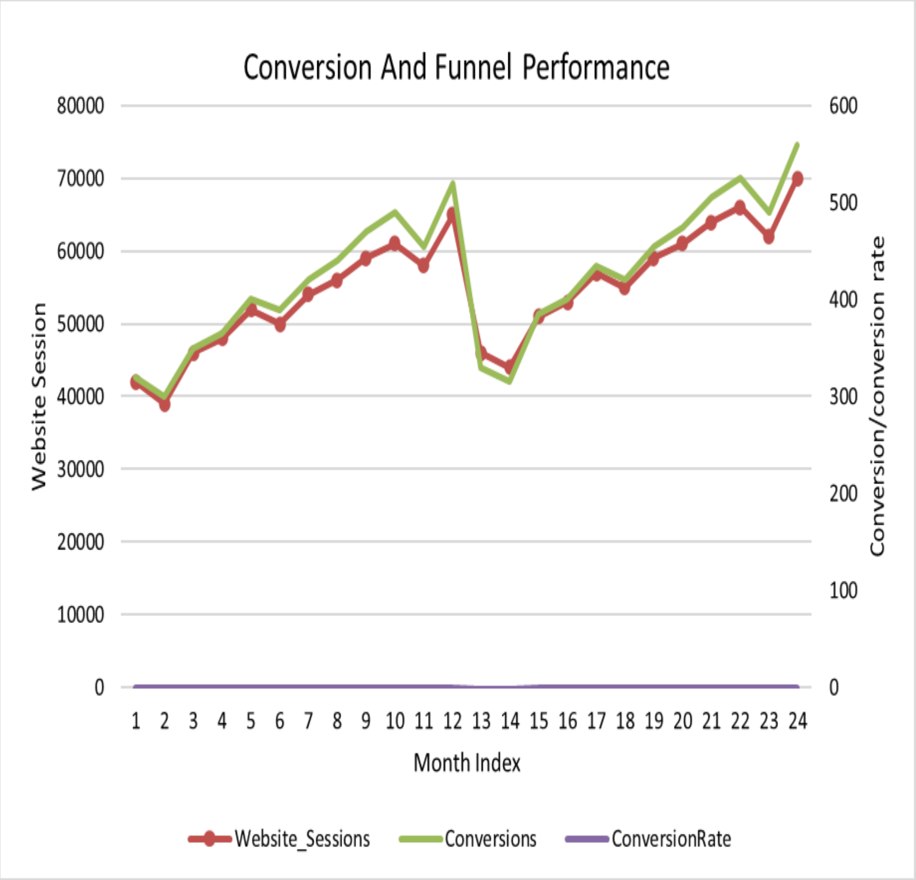
Revenue Trends

- Revenue shows a consistent upward trend over time
- Clear seasonal spikes appear in Q4 of each year
- Linear forecasts underestimate peak seasonal months compared to ETS



Conversions & Funnel Performance

- Conversion growth closely follows session growth
- Conversion rates remain relatively stable, enabling reliable conversion forecasts
- Forecasting conversions based on traffic assumptions proved more robust than trend-only models

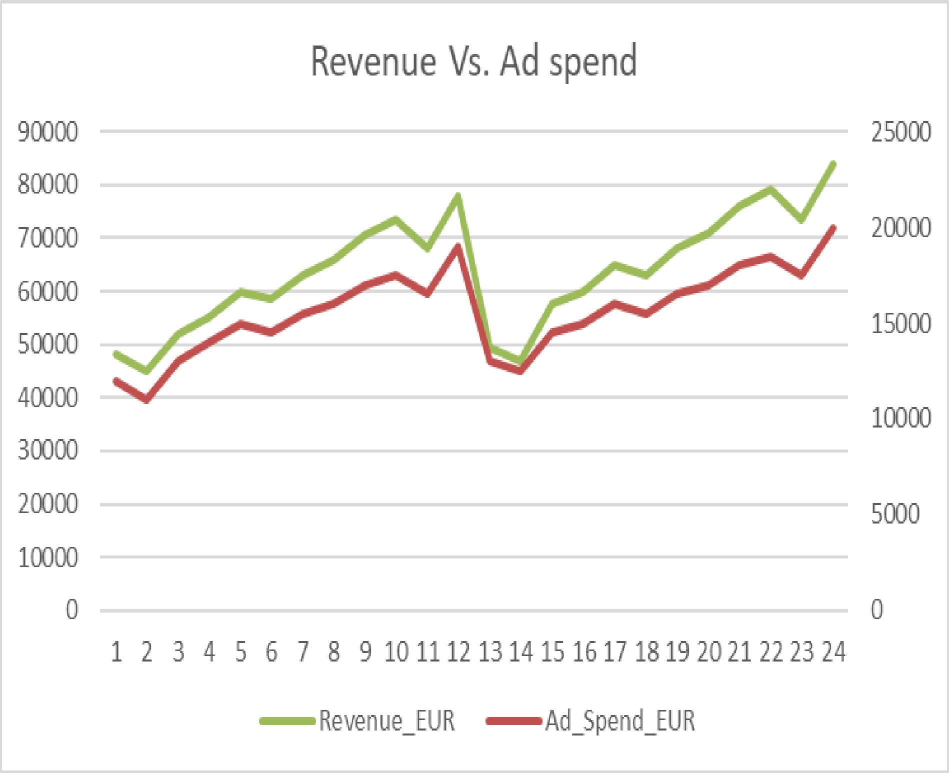


Ad Spend vs Revenue

Revenue increases with ad spend, but not proportionally

Signs of diminishing returns at higher spend levels

Indicates opportunities for efficiency optimization rather than pure budget increases



Business Impact

This analysis enables marketing stakeholders to:

- Set data-driven revenue and conversion targets
- Plan budgets with realistic performance expectations
- Anticipate seasonal demand and allocate resources accordingly
- Monitor forecast accuracy and adjust strategy proactively

Assumptions & Limitations

- Historical trends and seasonality continue into the forecast period
- No major changes in channel mix or market conditions
- Forecasts should be reviewed regularly as new data becomes available

Business Recommendations

Based on the forecasting analysis, the following actions are recommended to maximize marketing performance and planning accuracy:

1. Align Budgets With Seasonal Demand

- Increase marketing budgets ahead of historically strong Q4 periods to capture peak demand
- Reduce or optimize spend during lower-performing months to protect ROI

2. Prioritize Efficiency Over Spend Growth

- Revenue growth does not scale linearly with ad spend, indicating diminishing returns
- Focus on improving CPA and conversion rates through creative optimization, targeting, and landing page improvements rather than purely increasing budgets

Business Recommendations (cont.)

3. Use Multiple Forecasts for Decision-Making

- Apply linear trend forecasts for long-term planning and target setting
- Use moving averages and ETS forecasts for short-term execution and tactical adjustments

4. Review and Update Forecasts Regularly

- Re-forecast monthly as new data becomes available
- Adjust assumptions when major changes occur (campaign mix, pricing, market conditions)

5. Embed Forecasts Into Stakeholder Reporting

- Integrate forecasts into dashboards used by marketing and leadership teams
- Clearly communicate assumptions and confidence ranges to support informed decision-making

Tools Used

- Microsoft Excel
- Forecast.LINEAR
- Moving Averages
- Excel Forecast Sheet (ETS)
- Data visualization with line charts
- Combo charts

Key Takeaway

Forecasting delivers the most value when it informs concrete business actions. By combining trend-based and seasonality-aware forecasts with clear recommendations, marketing teams can plan more effectively, allocate budgets efficiently, and respond proactively to performance changes.

Marketing Revenue and Conversion Forcasting (Excel)

Total Revenue	Total Ad Spend	Marketing Return on Investment	Total Leads	Total Conversions	Website session
£1,531,000	£375,000	308.3%	65,700	10,215	1,318,000